Syafik Barda

Computer Engineering Student | Full-Stack & IoT Developer
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SUMMARY

Computer Engineering undergraduate at Diponegoro University (GPA 3.85/4.00) with a portfolio spanning full-stack web platforms, IoT telemetry systems, and applied machine learning. Experienced shipping production systems for both an industry client and university organizations, from Laravel and Next.js applications to solar-powered sensor nodes. Currently Chief of the Organizational Management and Development Unit at HIMASKOM, responsible for overseeing and evaluating the performance of the entire student association. Combining technical depth with organizational leadership, I am committed to delivering impactful contributions in dynamic engineering environments.

EXPERIENCE

PT WHS Maritime Investments South Jakarta, Indonesia
Website Developer *Jun 2025 – Aug 2025*

- Revamped the company digital presence by building a modern, performant company profile website aligned with PT WHS Maritime Investments brand identity.
- Collaborated with stakeholders to structure the information architecture, ensuring core services and business values are communicated clearly to potential clients.
- Engineered a Content Management System so the internal team can publish news, portfolio entries, and company information without developer involvement.

Tech: Laravel, PHP, React, Vite, Material UI, Bootstrap

PT SidoAgung Farm Magelang, Indonesia
IT Support *Jan 2022 – Jun 2022*

- Provided technical support to factory staff on network connectivity, hardware faults, and software configuration to keep daily operations running.
- Assisted in the installation and configuration of new network nodes and IT equipment during the expansion of factory operational areas.
- Monitored network traffic and performance, identifying and resolving bottlenecks to ensure fast and reliable data access across departments.

ORGANIZATIONAL EXPERIENCE

HIMASKOM – Diponegoro University Semarang, Indonesia
Chief of Organizational Management and Development Unit *Mar 2024 – Present*
Staff of Organizational Management and Development Unit *Sep 2023 – Mar 2024*

- Implemented a structural reform by establishing the Monitoring and Upgrading Division to oversee and raise the performance of all HIMASKOM 25 members.
- Served as Head of the Evaluation and Achievement Division, assessing work programs across every field and unit within HIMASKOM 24.
- Supervised and improved the performance of MIKATAN HIMASKOM 24, mentoring division heads and streamlining cross-unit coordination.

Computer Engineering Research Club (CERC) – Diponegoro University Semarang, Indonesia
Junior Staff, Software Division *Dec 2025 – Present*

- Study emerging software architectures, web protocols, and modern development frameworks together with the division across a one-year research program.

UNDIP Robotic Development Center (URDC) Semarang, Indonesia
Staff, Andromeda Division *Sep 2023 – Mar 2024*

- Completed a mentorship program focused on Search and Rescue (SAR) robot research, covering autonomous sensor integration and hardware-software control loops.

PROJECTS

Drift – Web-Based Animation Template Platform 2026 – Present

- Building a motion-template editor for people with no motion-design background: pick a Lottie preset, drop in your own text and logo, arrange the scene on a canvas, and export a showcase video without writing any animation code.
- Built the full stack on the Next.js App Router with the API as Route Handlers in the same codebase, plus end-to-end account handling: registration, email OTP verification, JWT sessions, and hashed credentials against PostgreSQL.

Tech: Next.js, React, PostgreSQL, JWT Auth, Lottie, GSAP, Tailwind CSS

CREATION 2026 Exam – Online Examination Platform with Proctoring 2026

- Developed a web-based examination platform for the CREATION 2026 event with continuous webcam proctoring over LiveKit, keeping a live view of every participant for the whole session.
- Implemented focus-lockdown detection that logs each tab and window switch to the database and surfaces it to proctors as a live violations feed, across a Laravel portal, an Express API, and a Next.js exam client joined by HMAC-SHA256 signed single sign-on.

Tech: Laravel, Next.js, Express.js, LiveKit, SSE, MySQL

SITIGA – RFID-Based Child Growth Monitoring Platform 2026

- Built a growth-monitoring platform for parents and health workers: a child taps an RFID card at the post, the measurement is recorded against WHO growth standards, and the history stays visible only to those permitted to see it.
- Implemented the RFID pipeline end to end (tag assignment, device-facing scan endpoint, scan logs) with API-key device authentication and role-based access across user, admin, and super-admin, including an approvals queue and bulk spreadsheet import.

Tech: Next.js, Express.js, MySQL, JWT Auth, RFID, Recharts, Tailwind CSS

Rat Pest Detection – Thermal Imaging with Machine Learning 2026

- Trained a Convolutional Neural Network on MLX90640 thermal captures to separate rat activity from background heat at night, driving ultrasonic repellers in rice paddies automatically.
- Built the field side on ESP32 and Raspberry Pi with PIR motion sensing, relay-driven emitters, and solar power, reporting telemetry to a dashboard covering device health, detection trends, and hot-spot ranking.

Tech: Python, CNN, Thermal Vision, ESP32 / C++, Raspberry Pi, Next.js, PostgreSQL

Swadaya Panen – IoT-Based Rice Field Monitoring System 2025

- Designed an integrated IoT system for precision farming, engineering multi-sensor field nodes that log soil moisture, air temperature, humidity, and light intensity on a single timestamped reading and stream it to a central server.
- Built a responsive web dashboard with threshold alarms so farmers can monitor crop health remotely and be alerted the moment conditions turn critical.

Tech: IoT Microcontroller, Sensors, Laravel, React, Vite, Chakra UI, MySQL

NPK Monitoring – Precision Soil Nutrient Analyzer 2025

- Developed an IoT system measuring soil Nitrogen, Phosphorus, and Potassium levels, integrating industrial NPK probes with microcontroller firmware on a standalone solar array.
- Built a dashboard over hourly and daily nutrient averages plus harvest logging, letting farmers determine precise fertilizer requirements and avoid soil degradation from over-fertilization.

Tech: C++ / Embedded, NPK Probes, Solar / PLTS, Laravel, React, Material UI, MySQL

ExpenseTracker – Personal Finance Management System 2025

- Developed a cross-platform mobile app on Expo Router and React Native, backed by a separate Express API, to help users track and categorize personal cash flow.
- Implemented dynamic charting of monthly cash flow and category breakdowns with PDF export, plus on-device caching so the ledger stays readable offline.

Tech: React Native, Expo, TypeScript, Supabase, Express.js, Chart Kit

HIMASKOM Company Profile Website 2024

- Designed and developed the official website for HIMASKOM UNDIP, presenting the association vision, work programs, division structure, and activities to prospective and current members.
- Modelled the organizational hierarchy, article index, and merchandise store as one typed data module so internal staff can keep the site current with minimal technical effort.

Tech: Next.js, React, TypeScript, CSS Modules, GSAP, Lenis

LEADERSHIP AND CERTIFICATION

Latihan Keterampilan Manajemen Mahasiswa Pra-Dasar (LKMM-PD) 2023
Diponegoro University

Leadership Training Teknik Komputer 2023
Diponegoro University

EDUCATION

Diponegoro University Semarang, Indonesia
Bachelor of Computer Engineering – GPA: 3.85 / 4.00 *Aug 2023 – Present*

ADDITIONAL

- **Languages & Frameworks:** JavaScript, TypeScript, PHP, Python, C++, React, Next.js, React Native, Node.js / Express, Laravel, Tailwind CSS
- **Data & Infrastructure:** MySQL, PostgreSQL, Supabase, REST APIs, WebSockets / SSE, JWT authentication, Git
- **IoT & Machine Learning:** ESP32 / microcontroller firmware, sensor integration (NPK, soil moisture, thermal), Raspberry Pi, CNN / computer vision
- **Soft Skills:** Leadership, Communication, Initiative, Responsibility, Teamwork, Problem Solving, Critical Thinking
- **Languages:** Bahasa Indonesia (Native), English (Fluent)